INTERIM FEDERAL SPECIFICATION

VALVE, FLOAT; ROD, TANK FLOAT; AND FLOAT, VALVE (WATER CLOSET) (LAND USE)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers float valves, tank float rods, and tank floats for standard size water closet tanks.

1.1.1 Federal Specification coverage. Federal Specifications do not include all varieties of float valves that are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Definitive part number (DPN). Items furnished under this specification shall be designated by a definitive part number. This number shall consist of the specification number followed by the specification part number (SPN) in table I. This part number is intended for cataloging and ordering purposes, and not for surface printing on parts (see 6.2 and 6.3). The DPN shall be written as indicated below:

DPN (see 6.3)---WWV1915 - XX
Federal Specification Number---
SPN (see 1.2.2)---

1.2.2 Specification part number identification. The type float valve and float valve parts shall be identified by the SPN listed in table I.

TABLE I. SPN identification

SPN	Item identification	Par.
A1	Type I float valve, separate float 1/	3.4
A2	Type II float valve, integral float	3.5
R1	Rod, tank float, 8" length	3.7.1
R2	Rod, tank float, 10" length	3.7.1
R3	Rod, tank float, 12" length	3.7.1
F1	Float, tank, copper	3.8.1
F2	Float, tank plastic	3.8.2

1/ Valve furnished without float

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28, Part I - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

American Society for Testing and Materials (ASTM) Standards:

B134 - Brass Wire, Specification for
D570 - Test for Water Absorption of Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 General. The materials specified shall conform to the applicable specifications and requirements specified hereinafter. Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice.

3.1.2 Plastics. Except as otherwise specified hereinafter, plastic parts continually immersed in water shall not absorb water in amounts greater than 0.42 percent by weight when tested in accordance with 4.4.1.

3.2 Standard product. The float valves and component parts covered by this specification shall be the manufacturer's standard commercial product, except as otherwise specified herein.

3.3 Construction. Float valves covered by this specification shall be antisiphon type. They shall be designed for use in a standard size water closet tank and shall be provided with a refill tube. The float valve shall be constructed so that it can be taken apart readily for repair. The float valve inlet shall provide for connection to a 3/8-inch supply tube. The supply boss or shank at the bottom of the valve assembly shall be brass or plastic.

3.4 Type I float valve, separate float (SPN-A1). Type I float valves shall be of brass casting with extensions of brass or copper tubing. All metal working parts shall be of copper alloy. Valve seats and parts connected to the valve outlet, so as not to contain water under pressure when the valve is closed, may be made of suitable non-metallic material that can be continuously submerged in water. The float valve shall be furnished with an 8-inch tank float rod in accordance with 3.7. The receptor for the tank float rod shall be threaded with .250-20 UNC 2B threads in accordance with H28. The length of the threads shall be not less than 1/2-inch.

3.5 Type II float valve, integral float (SPN-A2). Type II float valve body and float shall be made of copper, copper alloy, plastic, or combination thereof. The float valve shall have a 3-position (off-on-off) valve. The valve shall be so designed that water flowing into the tank shall be cut off when the valve float is at either the upper or lower limit of its travel. Water shall be permitted to flow into the tank when the float is at any other point between the two extremes. Means shall be provided by a secondary float to prevent the valve float from travel to its lower limit under normal operating conditions. Secondary floats of plastic shall be exempt testing in accordance with 3.1.2. Secondary floats will remain functional when tested as in accordance with 4.4.2 and 4.4.3.

3.6 Performance.

3.6.1 Water pressure. Float valves shall open and close effectively against water pressures up to 100 p.s.i.g. (see 4.4.2).

3.6.2 Back flow. There shall be no back flow from the tank into the float valve or supply tube when tested in accordance with 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.

3.7 Tank float rods (SPN-R1, R2, and R3). Tank float rods shall be of any copper alloy in accordance with ASTM B134. The temper of the rods shall be 1/8 hard or 1/4 hard. The tolerance on rod length shall be plus or minus 1/8-inch.

3.7.1 Rod dimensions. Rods shall be 1/4-inch nominal diameter. Ends of rods shall be threaded .250-20 UNC-2A in accordance with H28. The length of thread shall be 1 inch or greater at each end. The combined length of thread shall be not less than 3 inches. Length of rods shall be 8, 10, or 12 inches as specified by the SPN (see 1.2.2).

3.7.2 Bending property. Rods shall be capable of being bent through a 120 degree angle without fracture on the outside of the bend (see 4.4.5).

3.8 Tank float. Tank floats shall be copper or plastic as specified by the SPN (see 1.2.2).

3.8.1 Copper (SPN-F1). Copper floats shall be oval in shape, 4 inches by 5 inches and shall be formed in two halves with strip or cut sheet not less than 0.008 inch thick. Joint between halves shall be double reinforced crimped seam. Float shall be threaded to fit float rod (wire) having .250-20 UNC-2A threads in accordance with H28. Floats shall not leak when tested in accordance with 4.4.6.

3.8.2 Plastic (SPN-F2). Plastic floats shall be polyethylene (PE) or polystyrene (PS). Floats shall be ribbed wall construction and shall be hollow or of cellular or foam structure. Shape of floats shall be oval or ellipsoidal with a minor diameter of 4 inches and a major diameter of 6-1/4 inches. Tolerance on diameters shall be pluss or minus 1/4 inch. Floats shall be self-threaded. Color of floats shall be at the manufacturer's option. Hollow floats shall not leak when tested in accordance with 4.4.6.

3.9 Furnished items, types I and II float valves. Types I and II float valves shall be furnished with a ball cock gasket, a locknut, a slip joint washer, a friction ring, and a coupling nut.

3.10 Workmanship. All parts shall be free of cracks, dents, or other defects which affect serviceability or operation. All plastic parts shall be free of mold defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. A lot shall consist of all items with the same DPN offered for delivery at one time.

4.2.2 Sampling for examination and tests. Select sample size for examination and tests in accordance with level S-2 in MIL-STD-105.

4.3 Examination and tests.

4.3.1 Examination. Examine float valves, tank float rods, and float bulbs, as applicable for the defects listed in table II. The acceptable quality levels (AQL) for defects shall be 2.5 percent defective for major defects and 6.5 percent defective for total defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects, end item

Defects	Major	Minor
Item not in accordance with definitive part number	x	
Material not as specified.	x	
Thread size not as specified.	x	
Thread length not as specified.		x
Rod length not as specified.		x
Tank float dimensions not as specified		x
Metal and plastic fittings contain cracks.	x	
Mold defects on plastics		x
Damage or defects affecting serviceability	x	
Damage or defects not affecting serviceability		x

4.3.2 Test AQL. The AQL for all tests shall be 4.0 percent defective.

4.4 Test methods.

4.4.1 Water absorption, plastics. Use a part of a prepared specimen and test for water absorption in accordance with the 24 hour immersion procedure in ASTM D 570 (see 3.1.2).

4.4.2 Water pressure. Install valve in a standard water closet tank or other suitable container equipped with a flush valve. The float valve shall be installed with a float in accordance with this specification. Fill and flush tank under normal conditions. Pressurize inlet to 20 pounds per square inch gage (p.s.i.g.); fill and flush tank. Observe valve for conformance to 3.6.1. Repeat operation at pressures of 80 p.s.i.g. and 100 p.s.i.g. Prior to testing type II float valves, condition plastic secondary floats as follows

> Immerse secondary float in accordance with the 24 hour immersion procedure in ASTM D 570. At the end of the 24 hour period, test float valve in accordance with above procedure for conformance to 3.6.1.

Failure to pass this test shall cause rejection of the lot.

4.4.3 Shut-off test, type II float valve. Upon completion of the water pressure test, test type II float valves as follows:

a. Shut-off water supply to float valve.
b. Remove all water from tank.
c. Insure that the valve float at the lowest limit of travel.
d. Adjust water pressure to 100 p.s.i.g. and turn on water to float valve.

Observe float valve remains in the off position and that no water enters the tank (see 3.5). Failure to pass this test shall cause rejection of the lot.

4.4.4 Vacuum test. Connect water supply tube to a vacuum tank of suitable size. Attach a water vacuum of at least 15 inches of mercury and sustain it for 30 seconds. Observe the manometer for reduction in pressure (see 3.6.2).

4.4.5 Rod bending test. Cold bend rods through an angle of 120 degrees. The radius of the bend shall be equal to the diameter of the specimen. Examine specimen for conformance to 3.7.2.

4.4.6 Tank float leakage. Submerge three tank floats in water at ambient room temperature of 68° F. ± 5° F. (20° C. + 2.8° C.) for 24 hours. At end of test period, examine float and determine if water has entered float (see 3.8.1 and 3.8.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with Table III. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size; location, sequence, or method of application.
Material	Any component missing, damaged or not conforming to the applicable specification or requirements stated herein.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents	Quantity per container is more or less than required. Gross weight exceeds requirements of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Twelve rods of like description or one float or one valve shall be packaged in a close-fitting box conforming to PPP-B-636, class weather resistant, style optional. Close and water-proof seal the boxes in accordance with the appendix to PPP-B-636.

5.1.2 Level B. Twelve rods of like description or one float or one valve shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665 or to PPP-B-676. Closure of the boxes shall be in accordance with the applicable box specification or appendix thereto.

5.1.3 Level C. Twelve rods of like description or one float or one valve shall be packaged to insure protection during shipment and safe delivery to destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twenty-four unit packages of rods of like description (288 rods) packaged as specified in 5.1 or 24 floats or 24 valves of like description packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. Close, waterproof seal and strap the boxes in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Twenty-four unit packages of rods of like description (288 rods) packaged as specified in 5.1 or 24 floats or 24 valves of like description packaged as specified in 5.1 shall be packed in close-fitting box conforming to PPP-B-636, class domestic. Closure of the boxes shall be in accordance with Method II of the appendix to PPP-B-636.

5.2.3 Level C. Rods, valves or floats packaged as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The float valves, separate rods and floats are intended for use in standard size water closet tanks.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Definitive part number (see 1.2.1).
(c) Selection of applicable packing and packaging levels (see 5.1 and 5.2).
(d) Whether marking shall be in accordance with 5.3.1 or 5.3.2 (see 5.3).

6.3 Example of definitive part number.

DPN----------WWV1915-A1
Fed. Spec.
Type I tank float valve

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:24Z
Check the source to verify that this is the current version before use.